WhatsApp AI Support Bot: monthly running cost

Estimate for Tussles Watches, prepared 20 October 2026. Prices checked online the same day. All amounts in US dollars; PKR at about Rs 280 per $1 (open market, 13 Oct 2026).

Bill	What it pays for	How it is charged	Share of the cost
Meta (WhatsApp)	Every reply the bot sends	Per delivered message, after 1,000 free a month	**The biggest part** once traffic grows
Groq (AI)	Understanding the message, writing the reply, voice to text	Per token (text) and per second of audio	Small: a fraction of a cent per message
AWS (server)	The computer that runs the bot and its database	Per hour the server is on (fixed)	Covered by new-account credits at the start

1. Meta: WhatsApp Business Platform

- Since **1 October 2026** Meta charges for **service messages** (the bot's normal replies) again, per **delivered** message.
- Each business phone number gets **1,000 free service messages per month** (reset monthly, unused ones do not carry over). The 1,001st and later are charged.
- **Messages customers send are free.** Only what the business sends counts, and **every message counts**: one chat with 20 bot replies = 20 messages; a picture is its own message.
- A payment method must be in Meta Billing Hub, otherwise replies stop being delivered after the free 1,000.
- The test number used during development costs nothing; charges start on the real business number.

Pakistan rate (USD per delivered message, from 1 Oct 2026)	Service (bot replies)	Utility	Authentication	Marketing
Rate	$0.0150 after 1,000 free	$0.0150	$0.0150	$0.0473

The bot only sends service replies. Utility/marketing apply only to templates the business sends first (not used by this bot).

2. Groq: the AI models this project uses

Job in the bot	Model	Groq price	Cost per customer message
Understand the message (router) and write the reply (composer)	openai/gpt-oss-120b	$0.15 per 1M input tokens, $0.60 per 1M output tokens	about **$0.0006** (measured ~1,800 tokens per message, plus room for retries)
Voice note to text	whisper-large-v3	$0.111 per hour of audio, at least 10 s billed per note	about **$0.0006** per 20-second voice note
(cheaper option for voice)	whisper-large-v3-turbo	$0.04 per hour of audio	about $0.0002 per 20-second note

Groq is pay-as-you-go: no fixed monthly fee, you pay for what is used. The paid (Developer) tier is needed for production rate limits.

3. AWS: the server (EC2) for a new account

- A **new AWS account** (created after 15 July 2025) gets **$100 credit** at sign-up and **up to $100 more** for simple onboarding tasks. The Free plan lasts **6 months or until the credit is used**, whichever comes first.
- On the Free plan only some server types are allowed: t3.micro, t3.small, t4g.micro, t4g.small, c7i-flex.large, m7i-flex.large. Upgrade to the Paid plan before it ends (AWS keeps data 90 days, then deletes it).
- Region: **Mumbai (ap-south-1)**, the closest to Pakistan. The bot needs about 2 to 4 GB of memory (database + embedding model + app).

Server (Mumbai)	CPU / memory	Server per month	+ public IP + 30 GB disk	Total per month	When
t3.small	2 vCPU / 2 GB	$16.35	$3.65 + about $2.70	**about $23**	Free plan (credits): start here, with swap
c7i-flex.large	2 vCPU / 4 GB	$61.90	$3.65 + about $2.70	about $68	Free plan, if 2 GB is not enough (uses credit fast)
t3.medium	2 vCPU / 4 GB	$32.70	$3.65 + about $2.70	**about $39**	Paid plan (after the credits)

With t3.small the $200 credit covers the whole 6-month Free plan (about $136). Data out to the internet: first 100 GB a month free (the bot uses far less).

4. Total cost per month by traffic

One bot reply per customer message, 30 days a month, 20% of messages are voice notes (20 s each), no pictures counted. **During the AWS credits** the server is free for you; **after** it, the t3.medium server (about $39) is added.

Customer messages per day	Bot replies per month	Meta (WhatsApp)	Groq text	Groq voice	Total while AWS credit lasts	Total after credit (incl. ~$39 server)	In PKR (after credit)
30	900	$0.00	$0.54	$0.11	**$0.65**	**$39.73**	about Rs 11,100
100	3,000	$30.00	$1.80	$0.37	**$32.17**	**$71.25**	about Rs 20,000
300	9,000	$120.00	$5.40	$1.11	**$126.51**	**$165.59**	about Rs 46,400
500	15,000	$210.00	$9.00	$1.85	**$220.85**	**$259.93**	about Rs 72,800
1,000	30,000	$435.00	$18.00	$3.70	**$456.70**	**$495.78**	about Rs 138,800

How the Meta column is worked out: (bot replies per month – 1,000 free) × $0.015. Example for 100 messages a day: (3,000 – 1,000) × $0.015 = $30.
Up to about 33 replies a day the WhatsApp part is free (1,000 a month). Above that, Meta is 90% or more of the bill; the AI (Groq) stays a few dollars.

5. What is not in the table

- **Pictures**: each product picture the bot sends is one more Meta message ($0.015 after the free 1,000). The bot sends a picture only when asked (1 by default, 3 when the customer asks for more).
- **Domain name** (about $10 to $15 a year) for the HTTPS address Meta sends messages to.
- **Handoff alerts** go by email, so they cost nothing. If they are sent by WhatsApp to staff instead, those are template messages and are charged.
- **LangSmith** (tracing) is optional; the free tier was used up by testing this month, so it should stay off in production or be a paid plan.
- Prices change. Meta's Pakistan rate is from two independent pricing articles that cite Meta's rate card; confirm it on Meta's own rate card before go-live.

6. How the bot keeps the cost down (already built)

- One short reply per question, not several messages; pictures only on request.
- Rate limits per customer (per minute, hour and day): spam cannot run up the Meta bill.
- A daily budget setting: at 80% the team gets an alert, at 100% the bot only hands chats to a person.
- Voice can switch to whisper-large-v3-turbo (about one third of the price) if the accuracy test allows it.

7. Decisions for the store owner

- Who pays the Meta bill, and add a payment method in Meta Billing Hub before go-live.
- Which number the bot uses (a new SIM is recommended; the current WhatsApp number would leave the phone app).
- Groq paid plan, and when to move AWS from the Free plan to the Paid plan.

Sources

• Meta, Pricing on the WhatsApp Business Platform: developers.facebook.com/documentation/business-messaging/whatsapp/pricing
• Pakistan rates: engagelab.com/blog/whatsapp-business-api-pricing (updated 15 Oct 2026); digiteazy.com/whatsapp-business-api-pricing-pakistan (29 Sep 2026)
• Groq: console.groq.com/docs/speech-to-text; morphllm.com/groq-pricing (13 Oct 2026)
• AWS: aws.amazon.com/about-aws/whats-new/2025/07/aws-free-tier-credits-month-free-plan; docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-free-tier-usage.html; doit.com/compute (Mumbai on-demand prices); computingforgeeks.com/aws-costs-explained-real-numbers (public IPv4, data out)